

Advanced Engineering Mathematics

Complex Numbers

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1 Complex Numbers

Key Concept

A **complex number** is written in rectangular form as

$$z = x + iy,$$

where $x, y \in \mathbb{R}$ and $i^2 = -1$.

The quantities

$$\operatorname{Re}(z) = x, \quad \operatorname{Im}(z) = y$$

are called the **real part** and **imaginary part** of z , respectively.

The **complex conjugate** of z is

$$z^* = x - iy,$$

and the **modulus** of z is

$$|z| = \sqrt{x^2 + y^2}.$$

Complex numbers extend the real number system in a natural way. In earlier engineering mathematics, variables were often real-valued. Now the independent or dependent variable may be complex. This is important because many engineering tools, especially the Laplace transform and the Z -transform, use complex variables.

A useful viewpoint is geometric: the number

$$z = x + iy$$

is represented by the point (x, y) in the complex plane. The horizontal axis is the real axis, and the vertical axis is the imaginary axis.

Key Concept

For

$$z_1 = x_1 + iy_1, \quad z_2 = x_2 + iy_2,$$

the basic operations are

$$z_1 + z_2 = (x_1 + x_2) + i(y_1 + y_2),$$

$$z_1 - z_2 = (x_1 - x_2) + i(y_1 - y_2),$$

$$z_1 z_2 = (x_1 x_2 - y_1 y_2) + i(x_1 y_2 + x_2 y_1),$$

and if $z_2 \neq 0$,

$$\frac{z_1}{z_2} = \frac{(x_1 + iy_1)(x_2 - iy_2)}{x_2^2 + y_2^2}.$$

Also,

$$zz^* = |z|^2 = x^2 + y^2, \quad \operatorname{Re}(z) = \frac{z + z^*}{2}, \quad \operatorname{Im}(z) = \frac{z - z^*}{2i}.$$

Motivation and Intuition

Why do we need complex numbers in engineering? A sinusoid such as

$$A \cos(\omega t + \phi)$$

can be handled much more efficiently through

$$Ae^{i(\omega t + \phi)}.$$

Instead of separating magnitude and phase repeatedly, complex numbers allow both to be carried at once in a single expression.

Geometric Interpretation

If $z = x + iy$, then:

- $|z|$ is the distance from the origin to the point z .
- z^* is the reflection of z across the real axis.
- Addition corresponds to vector addition in the plane.
- Multiplication has a deeper geometric meaning that becomes clearer in polar form.

Key Properties

Key Concept

Important modulus properties:

$$|z_1 z_2| = |z_1| |z_2|, \quad \left| \frac{z_1}{z_2} \right| = \frac{|z_1|}{|z_2|} \quad (z_2 \neq 0),$$

$$|z_1 + z_2| \leq |z_1| + |z_2|$$

(the triangle inequality).

These are frequently used in complex analysis, signal processing, and control.

Example

Example 1: Simplify a quotient

Simplify

$$\frac{3 - 2i}{-1 + i}.$$

Multiply numerator and denominator by the conjugate of the denominator:

$$\frac{3 - 2i}{-1 + i} \cdot \frac{-1 - i}{-1 - i} = \frac{(3 - 2i)(-1 - i)}{(-1 + i)(-1 - i)}.$$

Compute the numerator:

$$(3 - 2i)(-1 - i) = -3 - 3i + 2i + 2i^2.$$

Since $i^2 = -1$,

$$-3 - 3i + 2i + 2(-1) = -5 - i.$$

Compute the denominator:

$$(-1 + i)(-1 - i) = (-1)^2 - i^2 = 1 - (-1) = 2.$$

Hence,

$$\frac{3 - 2i}{-1 + i} = \frac{-5 - i}{2} = -\frac{5}{2} - \frac{1}{2}i.$$

Example

Example 2: Use the conjugate to find real and imaginary parts

Let

$$z = 4 - 3i.$$

Then

$$z^* = 4 + 3i.$$

Now,

$$\operatorname{Re}(z) = \frac{z + z^*}{2} = \frac{(4 - 3i) + (4 + 3i)}{2} = \frac{8}{2} = 4,$$

and

$$\operatorname{Im}(z) = \frac{z - z^*}{2i} = \frac{(4 - 3i) - (4 + 3i)}{2i} = \frac{-6i}{2i} = -3.$$

Engineering Note

In AC circuit analysis, impedance is written as a complex number:

$$Z = R + iX,$$

where R is resistance and X is reactance. This compact form allows algebraic handling of phase shifts and amplitudes in one framework.

Common Mistakes / Notes

- Do not confuse z^2 with $|z|^2$. In general,

$$z^2 = (x + iy)^2 = x^2 - y^2 + i2xy,$$

whereas

$$|z|^2 = x^2 + y^2.$$

- The imaginary part of $z = x + iy$ is y , not iy .
- When dividing complex numbers, always rationalize the denominator using the conjugate.

2 Polar Form

Key Concept

If

$$z = x + iy,$$

then we may also write it in polar form as

$$z = r(\cos \theta + i \sin \theta) = re^{i\theta},$$

where

$$r = |z| = \sqrt{x^2 + y^2}, \quad x = r \cos \theta, \quad y = r \sin \theta.$$

Here r is the **modulus** and θ is an **argument** of z .

Polar form is especially useful because multiplication, division, powers, and roots become much simpler than in rectangular form.

Motivation / Intuition

In rectangular form, multiplication mixes real and imaginary parts. In polar form, multiplication becomes:

$$r_1 e^{i\theta_1} \cdot r_2 e^{i\theta_2} = (r_1 r_2) e^{i(\theta_1 + \theta_2)}.$$

So multiplication means:

- multiply the magnitudes,
- add the angles.

That is exactly why polar form is powerful in engineering applications involving gain and phase.

Geometric Interpretation

The number $z = re^{i\theta}$ is the point whose distance from the origin is r and whose angle from the positive real axis is θ .

Because angles differing by multiples of 2π represent the same direction,

$$\theta, \quad \theta + 2\pi, \quad \theta + 4\pi, \dots$$

all correspond to the same complex number.

Key Concept

The argument is **not unique**. If θ is one argument of $z \neq 0$, then every argument is

$$\theta + 2k\pi, \quad k \in \mathbb{Z}.$$

A commonly chosen value is the **principal argument**, denoted $\text{Arg } z$.

Key Properties / Theorems

Key Concept

Euler's formula:

$$e^{i\theta} = \cos \theta + i \sin \theta.$$

De Moivre's theorem:

$$(re^{i\theta})^n = r^n e^{in\theta} = r^n (\cos(n\theta) + i \sin(n\theta)).$$

Key Concept

For nonzero complex numbers,

$$z_1 = r_1 e^{i\theta_1}, \quad z_2 = r_2 e^{i\theta_2},$$

we have

$$\begin{aligned} z_1 z_2 &= r_1 r_2 e^{i(\theta_1 + \theta_2)}, \\ \frac{z_1}{z_2} &= \frac{r_1}{r_2} e^{i(\theta_1 - \theta_2)}. \end{aligned}$$

Example

Example 1: Convert to polar form

Express

$$z = -\sqrt{6} - i\sqrt{2}$$

in polar form.

First compute the modulus:

$$r = \sqrt{(-\sqrt{6})^2 + (-\sqrt{2})^2} = \sqrt{6+2} = 2\sqrt{2}.$$

Now determine the argument. Since

$$\tan \theta = \frac{-\sqrt{2}}{-\sqrt{6}} = \frac{1}{\sqrt{3}},$$

the reference angle is

$$\frac{\pi}{6}.$$

Because z lies in the third quadrant, the correct argument is

$$\theta = \pi + \frac{\pi}{6} = \frac{7\pi}{6}.$$

Hence,

$$z = 2\sqrt{2} e^{i7\pi/6}.$$

Example

Example 2: Compute a power using De Moivre's theorem

Find

$$(1 + i)^4.$$

First convert $1 + i$ to polar form.

Its modulus is

$$r = \sqrt{1^2 + 1^2} = \sqrt{2},$$

and its argument is

$$\theta = \frac{\pi}{4}.$$

Thus,

$$1 + i = \sqrt{2} e^{i\pi/4}.$$

Raise to the fourth power:

$$(1 + i)^4 = \left(\sqrt{2} e^{i\pi/4}\right)^4 = (\sqrt{2})^4 e^{i\pi} = 4(-1) = -4.$$

Example

Example 3: Find cube roots

Find all cube roots of

$$-1 + i.$$

Write the number in polar form. Its modulus is

$$r = \sqrt{(-1)^2 + 1^2} = \sqrt{2},$$

and one argument is

$$\phi = \frac{3\pi}{4}.$$

So

$$-1 + i = \sqrt{2} e^{i3\pi/4}.$$

The cube roots are

$$z_k = r^{1/3} e^{i(\phi+2k\pi)/3}, \quad k = 0, 1, 2.$$

Since

$$r^{1/3} = (\sqrt{2})^{1/3} = 2^{1/6},$$

we obtain

$$z_k = 2^{1/6} e^{i(\frac{3\pi}{4}+2k\pi)/3}, \quad k = 0, 1, 2.$$

Thus:

$$z_0 = 2^{1/6} e^{i\pi/4}, \quad z_1 = 2^{1/6} e^{i11\pi/12}, \quad z_2 = 2^{1/6} e^{i19\pi/12}.$$

Engineering Note

In frequency-response analysis, a transfer function value is often written as

$$H(i\omega) = |H(i\omega)| e^{i\phi(\omega)}.$$

The modulus gives gain and the argument gives phase shift. This is one reason polar form is central in electrical and control engineering.

Common Mistakes / Notes

- Do not determine θ using only $\tan^{-1}(y/x)$ without checking the quadrant.
- Remember that the argument is multivalued:

$$\arg z = \theta + 2k\pi.$$

- For roots, do not stop at one answer. An n th root problem has n distinct roots.

3 Complex Variables

Key Concept

A **complex variable** is a variable of the form

$$z = x + iy,$$

where both x and y may vary independently over subsets of $\mathbb{R}$.

Unlike a real variable, which moves along a line, a complex variable moves in a plane. This makes the behavior of complex functions richer and also more restrictive when differentiability is required.

Motivation / Intuition

In real calculus, a variable changes along one direction. In complex analysis, a variable can approach a point from infinitely many directions. This extra geometric freedom is the reason the theory becomes both more subtle and more powerful.

Formal Definitions

Key Concept

A **neighborhood** of z_0 is a disk

$$\{z : |z - z_0| < \varepsilon\}, \quad \varepsilon > 0.$$

A **deleted neighborhood** of z_0 is

$$\{z : 0 < |z - z_0| < \varepsilon\}.$$

A set is **open** if every point in it has a neighborhood contained in the set.

A set is **connected** if any two points in it can be joined by a path lying entirely in the set.

A **domain** is an open, connected set.

These ideas are needed because analyticity is not defined at just one isolated point. It is defined in neighborhoods and domains.

Geometric Interpretation

- The set

$$|z| < 1$$

is the open unit disk.

- The set

$$|z| \leq 1$$

contains the boundary circle and is not open.

- The set

$$0 < |z| < 1$$

is punctured at the origin but is still connected.

Key Ideas

Key Concept

Complex analysis is local. To say a function behaves well at a point, we usually require good behavior not just at that point but in some neighborhood around it.

This is why concepts such as open sets and domains are not merely formal language. They are essential.

Example

Example 1: Identify a domain

Determine whether

$$D = \{z : |z| < 2\}$$

is a domain.

The set consists of all points inside the circle of radius 2 centered at the origin.

It is open because each interior point has a small disk around it that still lies inside D .

It is connected because any two points inside the disk can be joined by a line segment lying inside the disk.

Hence,

$$D$$

is a domain.

Example

Example 2: A set that is not a domain

Consider

$$S = \{z : |z| < 1\} \cup \{z : |z - 4| < 1\}.$$

This is the union of two disjoint open disks.

It is open, because each disk is open.

However, it is not connected: a point in the first disk cannot be joined to a point in the second disk by a path staying entirely inside S .

Therefore, S is not a domain.

Engineering Note

In engineering, domains arise naturally when discussing where a transfer function is valid, where a logarithm is single-valued, or where a conformal map can be used safely.

Common Mistakes / Notes

- A set can be open without being connected.
- A set can be connected even if it has a hole.
- Analyticity is always discussed with respect to a domain, not just isolated points.

4 Complex Functions

Key Concept

A **complex function** is a mapping

$$f : \mathbb{C} \rightarrow \mathbb{C}.$$

If

$$z = x + iy,$$

then a complex function can be written as

$$w = f(z) = u(x, y) + iv(x, y),$$

where u and v are real-valued functions of two real variables.

Motivation / Intuition

A real function maps a line to a line. A complex function maps a plane to a plane. That means a complex function can move, rotate, stretch, fold, or invert regions of the plane.

This is why visualizing complex functions often involves studying how curves and regions in the z -plane map into the w -plane.

Geometric Interpretation

Useful curves in the complex plane include:

$$z = x + iy_0 \quad (\text{horizontal line}),$$

$$z = x_0 + iy \quad (\text{vertical line}),$$

$$z = re^{i\theta_0}, \quad r \geq 0 \quad (\text{ray}),$$

$$|z - c| = R \quad (\text{circle}).$$

A complex mapping may turn these simple curves into other lines, circles, spirals, or more complicated shapes.

Key Properties and Observations

Key Concept

If

$$f(z) = a + bz,$$

then:

- adding a translates points,
- multiplying by a real b dilates the plane,
- multiplying by a complex number $b = \rho e^{i\phi}$ both dilates by ρ and rotates by ϕ .

Key Concept

If

$$f(z) = z^2,$$

then

$$z = re^{i\theta} \implies w = r^2 e^{i2\theta}.$$

So the magnitude is squared and the argument is doubled.

Key Concept

If

$$f(z) = \frac{1}{z},$$

then

$$z = re^{i\theta} \implies w = \frac{1}{r}e^{-i\theta}.$$

So magnitudes are inverted and angles change sign.

Example

Example 1: Translation and rotation

Consider

$$f(z) = 2e^{i\pi/6}z + 1.$$

Let

$$z = re^{i\theta}.$$

Then

$$f(z) = 2re^{i(\theta+\pi/6)} + 1.$$

Interpretation:

1. multiply distances from the origin by 2,
2. rotate every point by $\pi/6$,
3. then translate the entire image one unit to the right.

Example

Example 2: Mapping under $w = z^2$

Let

$$z = e^{i\theta}, \quad 0 \leq \theta \leq \pi.$$

These points trace the upper semicircle of the unit circle.

Under

$$w = z^2,$$

we get

$$w = e^{i2\theta}.$$

As θ goes from 0 to π , the angle 2θ goes from 0 to 2π .

Therefore, the upper semicircle maps onto the entire unit circle.

Example

Example 3: Write $1/z$ in rectangular components

Let

$$w = \frac{1}{z}, \quad z = x + iy.$$

Then

$$w = \frac{1}{x + iy} \cdot \frac{x - iy}{x - iy} = \frac{x - iy}{x^2 + y^2}.$$

Hence

$$u(x, y) = \frac{x}{x^2 + y^2}, \quad v(x, y) = -\frac{y}{x^2 + y^2}.$$

Engineering Note

Complex mappings are fundamental in engineering because they connect geometry with system behavior. For example, mappings between the s -plane and the z -plane are central in control systems and digital signal processing.

Common Mistakes / Notes

- A complex function is not just a pair of unrelated real functions; u and v are linked when differentiability is required.
- The map z^2 is not one-to-one on the whole plane.
- The map $1/z$ is undefined at $z = 0$ and has singular behavior there.

5 Complex Differentiation

Key Concept

Let $f : \mathbb{C} \rightarrow \mathbb{C}$. The derivative of f at z is defined by

$$f'(z) = \lim_{\Delta z \rightarrow 0} \frac{f(z + \Delta z) - f(z)}{\Delta z},$$

provided the limit exists and is independent of the direction of approach.

Motivation / Intuition

Although the definition resembles that of real calculus, the key difference is that Δz can approach 0 from infinitely many directions in the complex plane.

This makes complex differentiability much more restrictive than real differentiability.

Practical Differentiation Rules

In practice, once a function is analytic, differentiation proceeds exactly like in real calculus.

Key Concept

If f and g are analytic, then:

$$(f \pm g)' = f' \pm g',$$

$$(fg)' = f'g + fg',$$

$$\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}, \quad g \neq 0,$$

$$(f \circ g)' = (f' \circ g) g'.$$

Analytic Functions

Key Concept

A function f is **analytic** at z_0 if it is differentiable in a neighborhood of z_0 .

If f is analytic everywhere in $\mathbb{C}$, it is called **entire**.

Key Observations

- Polynomials are entire.
- Rational functions are analytic wherever their denominators are nonzero.
- Functions involving z^* are generally not analytic.

Example

Example 1: Differentiate a polynomial

Find the derivative of

$$f(z) = z^2 + 3z + 1.$$

Using standard rules:

$$f'(z) = 2z + 3.$$

Example

Example 2: Differentiate a rational function

Find

$$\frac{d}{dz} \left(\frac{1}{z} \right).$$

Rewrite:

$$\frac{1}{z} = z^{-1}.$$

Differentiate:

$$\frac{d}{dz}(z^{-1}) = -z^{-2} = -\frac{1}{z^2}, \quad z \neq 0.$$

Example

Example 3: Non-analytic function

Let

$$f(z) = |z|^2 = x^2 + y^2.$$

This function cannot be differentiated using standard complex rules and is not analytic (except trivially at $z = 0$).

Engineering Note

In engineering applications, once analyticity is established, differentiation becomes straightforward and behaves exactly like real calculus. The real challenge is determining whether a function is analytic—this is where the Cauchy–Riemann equations become essential.

Common Mistakes / Notes

- Do not apply differentiation rules unless the function is analytic.
- The presence of z^* usually indicates non-analytic behavior.
- Complex differentiation is stricter than real differentiation due to direction independence.

6 Complex Integration

Key Concept

If a contour C is parametrized by

$$z(t) = x(t) + iy(t), \quad a \leq t \leq b,$$

then

$$dz = z'(t) dt = (x'(t) + iy'(t)) dt,$$

and the contour integral of $f(z)$ along C is

$$\int_C f(z) dz = \int_a^b f(z(t)) z'(t) dt.$$

Motivation / Intuition

In real calculus, we integrate along an interval. In complex analysis, we integrate along a path in the complex plane. Thus, both the function and the path matter.

For nonanalytic functions, the value may depend on the chosen path. For analytic functions in suitable domains, a remarkable simplification occurs: the integral becomes path independent.

Geometric / Physical Interpretation

You may think of a contour integral as accumulating the values of a complex field along a directed path. Orientation matters. Reversing the direction changes the sign of the integral.

Key Properties / Theorems

Key Concept

If f is analytic on and inside a simple closed contour C , then

$$\oint_C f(z) dz = 0.$$

This is the **Cauchy–Goursat theorem**.

Key Concept

If f is analytic in a domain, then integrals between two points in that domain are **path independent**. In particular, if $F'(z) = f(z)$, then

$$\int_a^b f(z) dz = F(b) - F(a).$$

This is the complex version of the fundamental theorem of calculus.

Worked Examples

Example

Example 1: Direct contour integration

Evaluate

$$\int_C z^2 dz$$

where C is the line segment from 0 to $2 + i$ given by

$$z(t) = (2 + i)t, \quad 0 \leq t \leq 1.$$

Differentiate:

$$dz = (2 + i) dt.$$

Also,

$$z^2(t) = ((2 + i)t)^2 = (2 + i)^2 t^2 = (3 + 4i)t^2.$$

Therefore,

$$\int_C z^2 dz = \int_0^1 (3 + 4i)t^2(2 + i) dt.$$

Multiply:

$$(3 + 4i)(2 + i) = 6 + 3i + 8i + 4i^2 = 2 + 11i.$$

So

$$\int_C z^2 dz = \int_0^1 (2 + 11i)t^2 dt = (2 + 11i) \int_0^1 t^2 dt = (2 + 11i) \left[\frac{t^3}{3} \right]_0^1.$$

Hence,

$$\int_C z^2 dz = \frac{2}{3} + \frac{11}{3}i.$$

Example

Example 2: Use an antiderivative

Evaluate

$$\int_C z^2 dz$$

for any contour from 0 to $2 + i$.

Since

$$f(z) = z^2$$

is entire, an antiderivative is

$$F(z) = \frac{z^3}{3}.$$

Therefore,

$$\int_C z^2 dz = F(2 + i) - F(0) = \frac{(2 + i)^3}{3}.$$

Compute:

$$(2 + i)^2 = 3 + 4i,$$

$$(2 + i)^3 = (3 + 4i)(2 + i) = 2 + 11i.$$

Hence,

$$\int_C z^2 dz = \frac{2 + 11i}{3} = \frac{2}{3} + \frac{11}{3}i.$$

The answer is the same for every contour joining the same endpoints, because z^2 is analytic everywhere.

Example

Example 3: Integral of $1/z$ around a circle

Evaluate

$$\oint_{|z|=r} \frac{1}{z} dz$$

counterclockwise.

Parametrize the circle by

$$z = re^{i\theta}, \quad 0 \leq \theta \leq 2\pi.$$

Then

$$dz = ire^{i\theta} d\theta.$$

Substitute:

$$\oint_{|z|=r} \frac{1}{z} dz = \int_0^{2\pi} \frac{1}{re^{i\theta}} (ire^{i\theta}) d\theta = i \int_0^{2\pi} d\theta = 2\pi i.$$

This is a foundational result in complex analysis.

Engineering Note

Contour integration is one of the most useful tools in engineering mathematics. It is used to evaluate inverse Laplace transforms, compute real integrals, analyze poles of transfer functions, and derive frequency-domain results.

Common Mistakes / Notes

- Always parametrize the contour carefully and compute dz correctly.
- Path independence requires analyticity on a suitable domain.
- The integral of $1/z$ around a closed contour enclosing the origin is not zero because $1/z$ is not analytic at $z = 0$.

7 Cauchy–Riemann Equations

Key Concept

Let

$$f(z) = u(x, y) + iv(x, y), \quad z = x + iy.$$

If f is differentiable at a point, then

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}.$$

These are the **Cauchy–Riemann equations**.

Motivation / Intuition

Because the derivative in the complex plane must be independent of direction, differentiating in the x -direction and in the y -direction must give the same result. That compatibility condition is exactly what produces the Cauchy–Riemann equations.

Derivation

Starting from

$$f'(z) = \lim_{\Delta z \rightarrow 0} \frac{\Delta w}{\Delta z},$$

take $\Delta z = \Delta x$ first. Then

$$f'(z) = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x}.$$

Now take $\Delta z = i\Delta y$. Then

$$f'(z) = \frac{\partial v}{\partial y} - i \frac{\partial u}{\partial y}.$$

Since both expressions must be equal,

$$\frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} = \frac{\partial v}{\partial y} - i \frac{\partial u}{\partial y}.$$

Equating real and imaginary parts gives

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \quad \frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y}.$$

Key Concept

If the partial derivatives are continuous in a neighborhood and the Cauchy–Riemann equations hold there, then f is differentiable there.

Polar Form

If

$$f(z) = u(r, \theta) + iv(r, \theta),$$

then the Cauchy–Riemann equations in polar coordinates are

$$\frac{\partial u}{\partial r} = \frac{1}{r} \frac{\partial v}{\partial \theta}, \quad \frac{\partial v}{\partial r} = -\frac{1}{r} \frac{\partial u}{\partial \theta}.$$

These are useful when the function is naturally expressed in polar form.

Worked Examples

Example

Example 1: Verify analyticity of $f(z) = z^2$

Write

$$z^2 = (x + iy)^2 = (x^2 - y^2) + i(2xy).$$

So

$$u(x, y) = x^2 - y^2, \quad v(x, y) = 2xy.$$

Now compute the partial derivatives:

$$u_x = 2x, \quad u_y = -2y, \quad v_x = 2y, \quad v_y = 2x.$$

Check the Cauchy–Riemann equations:

$$u_x = v_y \quad \Rightarrow \quad 2x = 2x,$$

$$u_y = -v_x \quad \Rightarrow \quad -2y = -2y.$$

They hold everywhere, and the partial derivatives are continuous everywhere. Therefore, z^2 is entire.

The derivative is

$$f'(z) = u_x + iv_x = 2x + i2y = 2z.$$

Example

Example 2: Test $f(z) = |z|^2$

Write

$$|z|^2 = x^2 + y^2 + i0.$$

Thus

$$u = x^2 + y^2, \quad v = 0.$$

Compute:

$$u_x = 2x, \quad u_y = 2y, \quad v_x = 0, \quad v_y = 0.$$

The equations require

$$2x = 0, \quad 2y = 0.$$

So they hold only at $(0, 0)$. Hence $|z|^2$ is not analytic in any neighborhood of any point, and therefore is not analytic anywhere.

Example

Example 3: Use the polar form for $f(z) = 1/z$

Write

$$f(z) = \frac{1}{z} = \frac{1}{r} e^{-i\theta} = \frac{\cos \theta}{r} - i \frac{\sin \theta}{r}.$$

So

$$u(r, \theta) = \frac{\cos \theta}{r}, \quad v(r, \theta) = -\frac{\sin \theta}{r}.$$

Compute:

$$u_r = -\frac{\cos \theta}{r^2}, \quad v_\theta = -\frac{\cos \theta}{r},$$

hence

$$\frac{1}{r} v_\theta = -\frac{\cos \theta}{r^2} = u_r.$$

Also,

$$v_r = \frac{\sin \theta}{r^2}, \quad u_\theta = -\frac{\sin \theta}{r},$$

so

$$-\frac{1}{r} u_\theta = \frac{\sin \theta}{r^2} = v_r.$$

Thus the polar Cauchy–Riemann equations hold for $r \neq 0$. Therefore, $1/z$ is analytic for $z \neq 0$.

Engineering Note

The Cauchy–Riemann equations are among the most important tests in engineering complex analysis. They help detect whether a candidate field can behave like a valid complex potential, analytic transfer function, or conformal mapping.

Common Mistakes / Notes

- Satisfying the Cauchy–Riemann equations at a single point does not guarantee analyticity there.
- Continuity of the partial derivatives in a neighborhood is the standard practical condition used with the equations.
- The equations test complex differentiability, not ordinary differentiability in $\mathbb{R}^2$.

8 Exponential and Logarithmic Functions

8.1 The Complex Exponential

Key Concept

The complex exponential is defined by

$$e^z = e^{x+iy} = e^x(\cos y + i \sin y), \quad z = x + iy.$$

Motivation / Intuition

The real exponential controls growth and decay. The imaginary exponential controls rotation and oscillation. In the complex exponential, both effects occur simultaneously.

Key Properties

Key Concept

For all complex numbers z, w ,

$$e^{z+w} = e^z e^w.$$

Also,

$$e^{iy} = \cos y + i \sin y.$$

Key Concept

The function e^z is **entire**.

To verify analyticity, write

$$e^{x+iy} = e^x \cos y + ie^x \sin y.$$

Thus

$$u(x, y) = e^x \cos y, \quad v(x, y) = e^x \sin y.$$

Then

$$u_x = e^x \cos y = v_y, \quad u_y = -e^x \sin y = -v_x.$$

So the Cauchy–Riemann equations hold everywhere.

Example

Example 1: Evaluate a complex exponential

Compute

$$e^{2+i\pi/3}.$$

Using

$$e^{x+iy} = e^x (\cos y + i \sin y),$$

we get

$$e^{2+i\pi/3} = e^2 \left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right).$$

Now

$$\cos \frac{\pi}{3} = \frac{1}{2}, \quad \sin \frac{\pi}{3} = \frac{\sqrt{3}}{2}.$$

Therefore,

$$e^{2+i\pi/3} = e^2 \left(\frac{1}{2} + i \frac{\sqrt{3}}{2} \right).$$

8.2 The Complex Logarithm

Key Concept

The complex logarithm is defined implicitly by

$$z = e^{\log z}.$$

If

$$z = re^{i\theta},$$

then

$$\log z = \ln r + i\theta = \ln |z| + i \arg z.$$

Why the Logarithm is Different

Unlike the exponential, the logarithm is **multivalued**, because the argument is multivalued:

$$\arg z = \theta + 2k\pi, \quad k \in \mathbb{Z}.$$

Hence,

$$\log z = \ln |z| + i(\theta + 2k\pi), \quad k \in \mathbb{Z}.$$

So one complex number has infinitely many logarithms.

Key Concept

The **principal logarithm** is

$$\operatorname{Log} z = \ln |z| + i \operatorname{Arg} z,$$

where usually

$$-\pi < \operatorname{Arg} z \leq \pi.$$

Branch Cuts and Analyticity

To make the logarithm single-valued, we restrict the domain. For the principal branch, the standard branch cut is along the nonpositive real axis.

Thus $\operatorname{Log} z$ is analytic on

$$\mathbb{C} \setminus (-\infty, 0].$$

Key Concept

For the principal branch,

$$\frac{d}{dz} \operatorname{Log} z = \frac{1}{z},$$

provided z stays in the chosen branch domain.

Example

Example 2: Find all logarithms of 1, -1 , and i

For 1:

$$1 = e^{i2k\pi}, \quad k \in \mathbb{Z}.$$

Hence

$$\log 1 = i2k\pi.$$

For -1 :

$$-1 = e^{i(\pi+2k\pi)},$$

so

$$\log(-1) = i(\pi + 2k\pi).$$

For i :

$$i = e^{i(\pi/2+2k\pi)},$$

hence

$$\log i = i \left(\frac{\pi}{2} + 2k\pi \right).$$

Example

Example 3: Compute the principal logarithm of $-1 + i$

First write

$$-1 + i = \sqrt{2} e^{i3\pi/4}.$$

Because

$$\text{Arg}(-1 + i) = \frac{3\pi}{4},$$

the principal logarithm is

$$\text{Log}(-1 + i) = \ln(\sqrt{2}) + i\frac{3\pi}{4}.$$

Since

$$\ln(\sqrt{2}) = \frac{1}{2} \ln 2,$$

we obtain

$$\text{Log}(-1 + i) = \frac{1}{2} \ln 2 + i\frac{3\pi}{4}.$$

Functions Derived from the Logarithm

Once the logarithm is available, many inverse and power functions can be defined:

$$z^\alpha = e^{\alpha \log z}.$$

This includes fractional powers and roots.

For example,

$$z^{1/n} = e^{\frac{1}{n} \log z}.$$

Because $\log z$ is multivalued, roots are also multivalued unless a branch is chosen.

Engineering Note

The complex exponential is indispensable in engineering because it unifies sinusoidal motion, phasors, damping, and oscillation. The logarithm is equally important in inverse mappings, stability analysis, phase unwrapping, and defining powers such as z^α .

Common Mistakes / Notes

- Do not treat $\log z$ as single-valued unless a branch has been specified.

- The identity

$$\log(z_1 z_2) = \log z_1 + \log z_2$$

must be used carefully in the complex setting because branch choices matter.

- The logarithm is undefined at $z = 0$ and discontinuous across its branch cut.

Quiz: Complex Numbers and Complex Functions

Instructions. Answer all items completely. Show all necessary solutions, including algebraic simplification, branch choices, and justification of analyticity where needed.

1. Polar Form, Powers, and Roots

Let

$$z = -8 + 8\sqrt{3}i.$$

- (a) Express z in polar form $z = re^{i\theta}$ using the principal argument.
- (b) Find all complex numbers w such that

$$w^4 = z.$$

- (c) Plot or describe geometrically the location of the four roots in the complex plane.

2. Complex Mapping

Consider the mapping

$$w = f(z) = \frac{1}{z - i}.$$

Let C be the horizontal line

$$z = x + 2i, \quad -\infty < x < \infty.$$

Find the image of C in the w -plane and describe its geometry completely.

3. Analyticity and the Cauchy–Riemann Equations

Let

$$f(z) = u(x, y) + iv(x, y),$$

where

$$u(x, y) = x^3 - 3xy^2 + 2x, \quad v(x, y) = 3x^2y - y^3 + ay.$$

Determine the value of the real constant a such that f is analytic everywhere. Then express $f(z)$ purely in terms of z .

4. Complex Integration

Evaluate

$$\int_C \bar{z} dz$$

along each of the following paths from 0 to $1 + i$:

- (a) C_1 : the line segment $z(t) = (1 + i)t$, $0 \leq t \leq 1$;

(b) C_2 : the polygonal path from 0 to 1, then from 1 to $1 + i$.

Based on your answers, explain whether $\bar{z}$ has path-independent integrals.

5. Complex Exponential and Logarithm

Solve

$$e^z = -2 + 2i$$

for all complex numbers z . Then identify the principal solution using the principal argument.

Solutions: Complex Numbers and Complex Functions Quiz

1. Polar Form, Powers, and Roots

Given

$$z = -8 + 8\sqrt{3}i.$$

First compute the modulus:

$$r = \sqrt{(-8)^2 + (8\sqrt{3})^2} = \sqrt{64 + 192} = \sqrt{256} = 16.$$

Since $x = -8 < 0$ and $y = 8\sqrt{3} > 0$, the point lies in Quadrant II. Also,

$$\tan \theta = \frac{8\sqrt{3}}{-8} = -\sqrt{3}.$$

The reference angle is $\pi/3$, so the principal argument is

$$\theta = \frac{2\pi}{3}.$$

Hence,

$$z = 16e^{i2\pi/3}.$$

We now solve

$$w^4 = 16e^{i2\pi/3}.$$

Write

$$w = \rho e^{i\phi}.$$

Then

$$w^4 = \rho^4 e^{i4\phi}.$$

Therefore,

$$\rho^4 = 16, \quad 4\phi = \frac{2\pi}{3} + 2k\pi.$$

Thus,

$$\rho = 2,$$

and

$$\phi_k = \frac{1}{4} \left(\frac{2\pi}{3} + 2k\pi \right) = \frac{\pi}{6} + \frac{k\pi}{2}, \quad k = 0, 1, 2, 3.$$

Therefore, the four roots are

$$w_k = 2e^{i\left(\frac{\pi}{6} + \frac{k\pi}{2}\right)}, \quad k = 0, 1, 2, 3.$$

Explicitly:

$$\begin{aligned} w_0 &= 2e^{i\pi/6}, & w_1 &= 2e^{i2\pi/3}, \\ w_2 &= 2e^{i7\pi/6}, & w_3 &= 2e^{i5\pi/3}. \end{aligned}$$

Geometrically, the roots lie on the circle

$$|w| = 2$$

and are equally spaced by

$$\frac{2\pi}{4} = \frac{\pi}{2}.$$

Hence, they form the vertices of a square centered at the origin.

2. Complex Mapping

We are given

$$w = \frac{1}{z - i}$$

and

$$z = x + 2i.$$

Substitute into the mapping:

$$w = \frac{1}{x + 2i - i} = \frac{1}{x + i}.$$

Rationalize:

$$w = \frac{1}{x + i} \cdot \frac{x - i}{x - i} = \frac{x - i}{x^2 + 1}.$$

Write

$$w = u + iv.$$

Then

$$u = \frac{x}{x^2 + 1}, \quad v = -\frac{1}{x^2 + 1}.$$

We eliminate x . Divide u by v :

$$\frac{u}{v} = \frac{x/(x^2 + 1)}{-1/(x^2 + 1)} = -x.$$

Hence,

$$x = -\frac{u}{v}.$$

Since

$$v = -\frac{1}{x^2 + 1},$$

substitute $x = -u/v$:

$$v = -\frac{1}{(u^2/v^2) + 1}.$$

Simplify:

$$v = -\frac{v^2}{u^2 + v^2}.$$

Since $v \neq 0$ on the image curve, divide by v :

$$1 = -\frac{v}{u^2 + v^2}.$$

Therefore,

$$u^2 + v^2 = -v.$$

Rearranging:

$$u^2 + v^2 + v = 0.$$

Complete the square:

$$u^2 + \left(v + \frac{1}{2}\right)^2 = \frac{1}{4}.$$

Hence, the image is the circle

$$u^2 + \left(v + \frac{1}{2}\right)^2 = \left(\frac{1}{2}\right)^2.$$

This is a circle of radius $1/2$ centered at

$$0 - \frac{1}{2}i.$$

3. Analyticity and the Cauchy–Riemann Equations

We are given

$$u(x, y) = x^3 - 3xy^2 + 2x, \quad v(x, y) = 3x^2y - y^3 + ay.$$

Compute the partial derivatives:

$$u_x = 3x^2 - 3y^2 + 2, \quad u_y = -6xy,$$

$$v_x = 6xy, \quad v_y = 3x^2 - 3y^2 + a.$$

The Cauchy–Riemann equations require

$$u_x = v_y, \quad u_y = -v_x.$$

The second equation gives

$$-6xy = -6xy,$$

which is automatically satisfied.

The first equation gives

$$3x^2 - 3y^2 + 2 = 3x^2 - 3y^2 + a.$$

Therefore,

$$a = 2.$$

With $a = 2$,

$$v(x, y) = 3x^2y - y^3 + 2y.$$

Now observe that

$$z^3 = (x + iy)^3 = x^3 + 3x^2iy + 3x(iy)^2 + (iy)^3.$$

Since $i^2 = -1$ and $i^3 = -i$,

$$z^3 = x^3 + 3ix^2y - 3xy^2 - iy^3.$$

Hence,

$$z^3 = (x^3 - 3xy^2) + i(3x^2y - y^3).$$

Also,

$$2z = 2x + i2y.$$

Therefore,

$$z^3 + 2z = (x^3 - 3xy^2 + 2x) + i(3x^2y - y^3 + 2y).$$

Thus,

$$f(z) = z^3 + 2z.$$

4. Complex Integration

We evaluate

$$\int_C \bar{z} dz.$$

Path C_1 : Let

$$z(t) = (1 + i)t, \quad 0 \leq t \leq 1.$$

Then

$$\overline{z(t)} = (1 - i)t, \quad dz = (1 + i) dt.$$

Thus,

$$\int_{C_1} \bar{z} dz = \int_0^1 (1 - i)t(1 + i) dt.$$

Since

$$(1 - i)(1 + i) = 1 - i^2 = 2,$$

we get

$$\int_{C_1} \bar{z} dz = \int_0^1 2t dt = [t^2]_0^1 = 1.$$

Path C_2 : This path has two segments.

First segment: from 0 to 1,

$$z(t) = t, \quad 0 \leq t \leq 1.$$

Then

$$\bar{z} = t, \quad dz = dt.$$

Thus,

$$\int_0^1 \bar{z} dz = \int_0^1 t dt = \frac{1}{2}.$$

Second segment: from 1 to $1 + i$,

$$z(t) = 1 + it, \quad 0 \leq t \leq 1.$$

Then

$$\bar{z} = 1 - it, \quad dz = i dt.$$

Therefore,

$$\int (1 - it)i \, dt = \int (i + t) \, dt.$$

Hence,

$$\int_0^1 (i + t) \, dt = i + \frac{1}{2}.$$

Adding both segments:

$$\int_{C_2} \bar{z} \, dz = \frac{1}{2} + \left(i + \frac{1}{2}\right) = 1 + i.$$

Since

$$\int_{C_1} \bar{z} \, dz = 1$$

but

$$\int_{C_2} \bar{z} \, dz = 1 + i,$$

the integral depends on the path.

Therefore, $\bar{z}$ does not have path-independent integrals.

5. Complex Exponential and Logarithm

We solve

$$e^z = -2 + 2i.$$

First express the right-hand side in polar form:

$$-2 + 2i = 2\sqrt{2} e^{i3\pi/4}.$$

Since the argument is multivalued,

$$-2 + 2i = 2\sqrt{2} e^{i(3\pi/4 + 2k\pi)}, \quad k \in \mathbb{Z}.$$

Let

$$z = x + iy.$$

Then

$$e^z = e^{x+iy} = e^x e^{iy}.$$

Matching modulus and argument gives

$$e^x = 2\sqrt{2}, \quad y = \frac{3\pi}{4} + 2k\pi.$$

Therefore,

$$x = \ln(2\sqrt{2}).$$

Hence all solutions are

$$z = \ln(2\sqrt{2}) + i \left(\frac{3\pi}{4} + 2k\pi \right), \quad k \in \mathbb{Z}.$$

Since

$$\ln(2\sqrt{2}) = \ln(2^{3/2}) = \frac{3}{2} \ln 2,$$

we may also write

$$z = \frac{3}{2} \ln 2 + i \left(\frac{3\pi}{4} + 2k\pi \right), \quad k \in \mathbb{Z}.$$

The principal solution uses

$$\text{Arg}(-2 + 2i) = \frac{3\pi}{4}.$$

Therefore,

$$z_{\text{principal}} = \frac{3}{2} \ln 2 + i \frac{3\pi}{4}.$$